Re-Implementing CNN Price-Trend Signals: A Reproduction of Jiang, Kelly, and Xiu (2023)

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Abstract

This paper present my **final project** of the course **Machine Learning for Finance,** which reproduces the image-based price-trend prediction framework of Jiang et al. [2023] (hereafter the *original paper*), which replaces predefined technical indicators with OHLC-volume charts and convolutional neural networks (CNNs). Using CRSP daily U.S. equity data, I reconstruct the image pipeline, train horizon-specific CNN ensembles, and evaluate the resulting signals through out-of-sample backtests. I compare the CNN signals with momentum, moving-average, and reversal benchmarks, and assess residual performance through factor-spanning regressions. The reproduction is strongest at short horizons, where image-based signals deliver the best economic performance. Relative to the magnitudes reported by the original paper, medium- and long-horizon strategies weaken substantially after costs, and factor stripping yields significantly negative alphas. Saliency maps indicate that the CNN mainly attends to recent price movement and volume contrast. I also provide replication materials so that the empirical results can be audited and extended.

1 Introduction

Technical analysis has long suggested that the visual "shape" of a price chart conveys information beyond what is captured by scalar momentum or moving-average filters. The original paper formalises this intuition by feeding fixed-resolution open–high–low–close (OHLC) images with a volume sub-panel into a convolutional neural network and showing that the resulting forecasts dominate canonical price-trend signals out of sample. The study joins a growing empirical asset-pricing literature that uses flexible machine-learning predictors to map raw historical features into expected returns [Gu et al., 2020, Kelly et al., 2019], and it ties directly to early efforts at formalising chart patterns via kernel smoothing [Lo et al., 2000].

This paper re-implements the core pipeline from end to end and evaluates it under the same in-sample and out-of-sample calendar design as the original paper: training and validation draw from 1993–2000 with a 70/30 random sample split, and headline out-of-sample results are reported for 2001–2019. My contributions are:

1. An independent OHLC-plus-volume image renderer and horizon-matched CNN stacks built from the dimensions, drawing rules, and layer geometry documented in the original paper—including the simple moving-average overlay, volume sub-panel, and vertical orientation convention—implemented from the original paper's written specification and reported tensor dimensions.
2. A CRSP-style daily panel for data construction, with headline out-of-sample CNN evaluation as above; proportional transaction costs; matched-horizon price momentum, simple moving-average cross, and short-term reversal baselines; and a five-factor-plus-momentum spanning regression [Fama and French, 1993, 2015, Carhart, 1997].

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3. Input-gradient saliency analysis [Simonyan et al., 2014] of the $I = 60$ ensemble, which highlights the right tail of each image and the volume bars as the dominant feature regions.
4. **Code, trained weights, and post-aggregation CSV tables are released on GitHub, allowing every numerical claim to be regenerated.**

The remainder of the paper is organised as follows. Section 2 positions my reproduction relative to the chart-pattern, deep-learning, and interpretability literatures. Section 3 details data sources and image construction. Section 4 describes the CNN architecture and training recipe. Section 5 reports empirical results: cross-sectional decile analyses against momentum, moving-average cross, and short-term reversal benchmarks; volatility-adjusted cumulative long–short performance (Figure 5–style); portfolio statistics; classification metrics; and Fama–French factor spanning. Section 6 visualises model attention. Section 7 discusses where my results align with and diverge from the original paper.

2 Related work

Chart-pattern literature. Lo et al. [2000] provided one of the first systematic treatments of technical chart patterns by combining kernel regression with automated pattern recognition. Jegadeesh and Titman [1993] documented price momentum as a robust cross-sectional return predictor, providing the canonical baseline against which chart-based technical trend signals are often benchmarked empirically.

Deep learning for return prediction. Recent work demonstrates that deep architectures uncover return signals that are difficult to capture with closed-form characteristics. Gu et al. [2020] benchmark a broad set of supervised learners on U.S. equities; Kelly et al. [2019] use latent-factor models with characteristic loadings. Within technical analysis, Sezer and Ozbayoglu [2018] convert indicators into images for trading. The original paper unifies these threads by treating the OHLC chart itself as the model input.

Interpretability via input gradients. Simonyan et al. [2014] introduced gradient-based saliency to visualise which input pixels most influence a CNN's class score; I apply the same technique to identify which segments of each OHLC image drive the model's "up" probability. CNN architectures used here trace back to LeCun et al. [1998].

3 Data and image construction

Universe. I employ a CRSP-consistent daily security-level panel of U.S. equities spanning 1992-01-01, to 2024-12-31. Data are observed at the daily frequency for each security, uniquely identified by its CRSP permanent security number (PERMNO). The panel includes core trading and pricing fields: daily market capitalization, total and price returns, trading volume, and open-high-low-close (OHLC) prices. I impose a standard sequential screening procedure: all fields are required to be non-missing; I retain only securities with a closing price of at least \$1 and strictly positive trading volume. Log returns are computed, and OHLC prices are adjusted for corporate actions via cumulative log-return compounding, with the first valid observation for each security serving as the normalization anchor. No additional survivorship bias adjustments are applied beyond those inherent to the standard CRSP universe construction.

Labels and holding period. For each horizon $F \in \{5, 20, 60\}$ trading days I form $R^{\text{fut}}_{t,F} = \exp\big(\sum_{s=t+1}^{t+F} \log(1 + r_s)\big) - 1$ on the adjusted price path, and a binary classification label $y_{t,F} = \mathbf{1}\{R^{\text{fut}}_{t,F} > 0\}$. Rebalance timestamps are fixed at period-end calendar boundaries (Friday for $F = 5$, month-end for $F = 20$, quarter-end for $F = 60$), matching the canonical SPY trade calendar.

Image construction. Following the original paper, I encode each stock–date observation as a fixed-size grayscale *chart image* rather than a vector of technical indicators. The background is black (pixel value 0) and all price–volume objects are drawn in white (255). I consider three input horizons $I \in \{5, 20, 60\}$, each matched to the prediction horizon $F = I$, so the image summarizes the most recent I trading days of adjusted OHLC prices and daily volume.

OHLC panel. The upper portion of the image is reserved for open–high–low–close prices. Each trading day τ in the I-day window occupies three adjacent pixel columns: a one-pixel open marker in the left column, a vertical high–low bar in the centre, and a one-pixel close marker in the right column. Prices are linearly mapped into the vertical axis using the window minimum and maximum, where the range is defined to include both the raw OHLC values and the moving-average line described below. After drawing, the OHLC sub-panel is flipped top-to-bottom so that upward price movements read visually as "up" on the chart, in line with conventional price plots.

Moving-average line. I overlay a simple moving average (SMA) of the close price, using an I-day window—the same length as the chart horizon (e.g., a 20-day chart uses a 20-day SMA). For each day, the SMA value is placed on the centre column of that day; consecutive days are connected by a one-pixel poly-line. To ensure that the first in-window SMA already reflects I days of history, the SMA is computed from $2I - 1$ days of past closes ending at the chart date.

Volume sub-panel. The lower portion of the image displays volume bars. As in the original paper, the volume chart is drawn along the bottom of the image and occupies one-fifth of the total height; the OHLC panel fills the upper four-fifths, with a one-pixel gap between the two regions. Within the volume band, the largest daily volume in the window is scaled to the top of the band and all other days are drawn proportionally. Missing-volume days are left blank.

Image dimensions. Because each day uses three pixel columns, the image width is $W = 3I$. The total height H stacks the OHLC block, the separator, and the volume band. Table 1 reports the (W, H) pairs used in my implementation, identical to those in the original paper.

Table 1: Image Dimensions

Horizon	W	H
$I = 5$	15	32
$I = 20$	60	64
$I = 60$	180	96

Figure 1 provides a concrete illustration of the construction rules above. I plot the same security on the same calendar anchor for $I \in \{5, 20, 60\}$, so the only deliberate difference across panels is the look-back length. The left panel ($I = 5$) is the coarsest view: fifteen columns of price action and a compact volume strip. The centre and right panels progressively widen the canvas to sixty and one hundred eighty columns, revealing more fine-grained OHLC structure, a longer SMA trace, and richer intra-window variation in volume. Because every image fed to the CNN has fixed (W, H) for a given horizon, longer windows do not add pixels per trading day; instead, the model sees the same economic history at three standardised resolutions, analogous to viewing a price chart at weekly, monthly, and quarterly zoom levels.

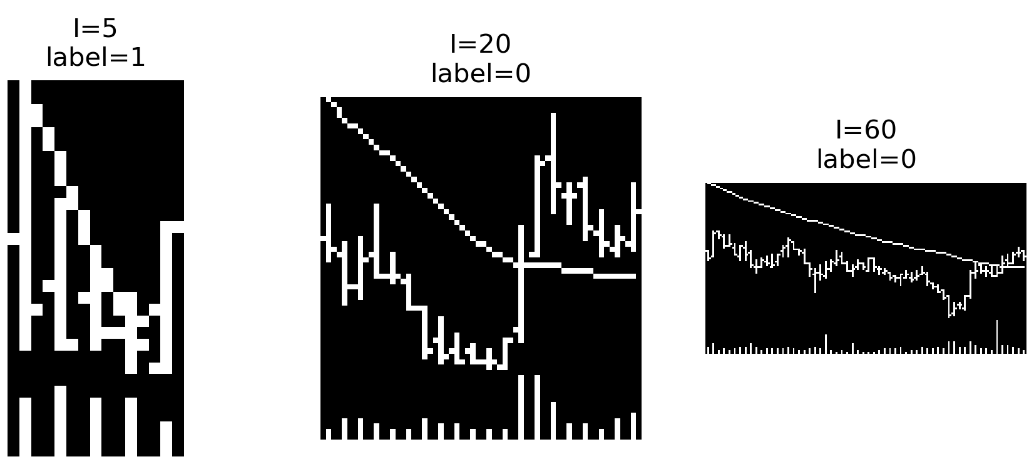


Figure 1: OHLC-plus-volume images for a single security at horizons $I \in \{5, 20, 60\}$.

Train/validation/test split. I follow the original paper in treating the early portion of the panel as in-sample. Training is performed on 1993–2000 with a 70/30 random sample-level split for training/validation. Out-of-sample backtests start in January 2001. Headline tables and figures report results through December 2019, matching the evaluation window in the original paper.

4 CNN architecture and training

Architectures. The three horizon-specific backbones mirror the published configuration. Each block applies a 5×3 convolution, LeakyReLU(0.01), batch normalisation, and 2×1 max pooling along the time axis. For $I = 20$ and $I = 60$, the opening convolution uses dilation and stride to widen the receptive field without a proportional increase in parameters, and pooling is configured so that spatial dimensions are preserved enough to yield the fully connected widths reported in the original paper; the $I = 5$ network is shallower and uses the default pooling rule throughout. Figure 2 summarises the three stacks; stride, dilation, and channel widths are listed in Table 5 (Appendix B).

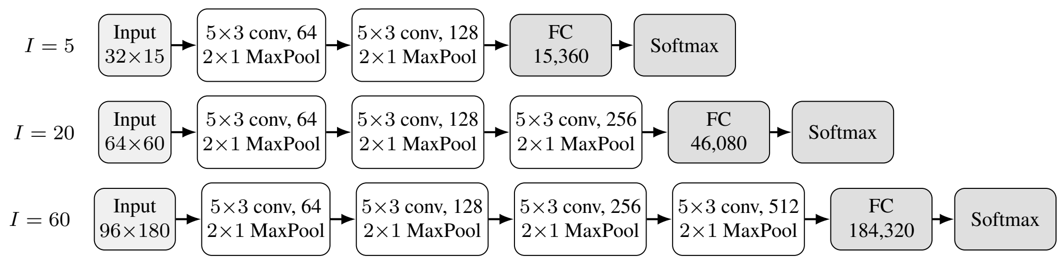


Figure 2: Horizon-specific CNN backbones. Each rectangle is one block: $5{\times}3$ convolution (with the channel width shown), LeakyReLU, batch normalisation, and $2{\times}1$ max pooling. Stride and dilation for $I = 20$ and $I = 60$ are in Table 5. Fully connected input sizes are 15,360 ($I = 5$), 46,080 ($I = 20$), and 184,320 ($I = 60$).

Optimisation. I minimise binary cross-entropy on the per-horizon label $y_{t,F}$ using stochastic gradient descent (Adam variant disabled), with learning rate 10^{-5}, batch size 128, and a maximum of 50 epochs with early-stopping on validation loss. Convolutional and linear weights are initialised with Xavier-uniform and biases at zero.

The original paper reports headline results from an ensemble of independently trained networks whose softmax outputs are averaged at inference. My replication proceeded in two stages. For the course presentation, limited local compute allowed only a single random seed (42) per horizon and I restricted the panel to the Top 1,000 stocks by market capitalization to keep training tractable, so I could not yet reproduce the full ensemble-averaging protocol or match the original cross-section breadth. For the final project I reran training on rented cloud GPUs on the full eligible universe (without Top 1,000 truncation). I use five independent seeds $\{42, 123, 456, 789, 101112\}$ and average softmax-probability outputs at inference time, as in the original paper. All empirical results in Sections 5–6 use this five-seed ensemble. The resulting model sizes are approximately 0.6 MB ($I = 5$), 2.8 MB ($I = 20$), and 11.8 MB ($I = 60$) per seed, consistent with the parameter counts implied by Figure 2.

5 Empirical results

I score every test-set observation with the five-seed ensemble and form ten cross-sectional deciles by predicted "up"-probability at each rebalance date. The high-minus-low (H–L) portfolio is equal-weighted within deciles, rebalanced at horizon-end, and adjusted for proportional transaction costs at 10 bp per side (20 bp round-trip) applied to both legs.

5.1 Cross-sectional deciles versus conventional benchmarks

My first empirical object is unconditional decile-sort performance. At each horizon $I \in \{5, 20, 60\}$ I partition stocks into predicted-probability deciles using the CNN ensemble score, and I form analogous deciles from matched-horizon momentum, moving-average crossover, and short-term reversal signals constructed on the same out-of-sample panel and rebalance calendar. For every horizon I summarise average annualised realised *levels* and volatilities by decile in the dual-panel layout of Figure 7 in the original paper (returns on the left, volatilities on the right), overlaying all four signals to contrast the CNN's ranking with traditional patterns. To keep the main exposition readable, Figure 3 reports only the $I = 5$ (weekly) case, which is also where the economic spread survives costs most clearly; Appendix D reproduces the identical format for $I = 20$ and $I = 60$.

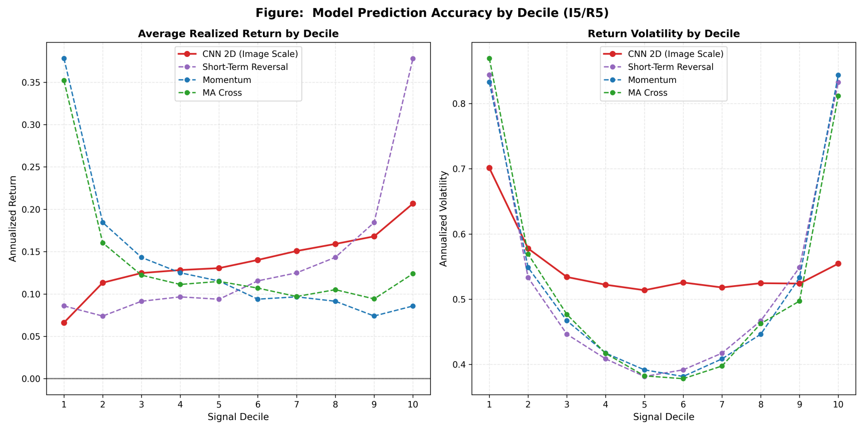


Figure 3: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 5$ horizon. Reproduction of Figure 7 of the original paper. The reversal baseline's D10 reaches an annualised return of 37.78%, against which the CNN's monotone ranking is more useful as a cross-sectional signal than as a stand-alone strategy. Analogous plots for $I = 20$ and $I = 60$ are in Appendix D.

At $I = 5$, the CNN traces a stable, nearly monotone gradient across deciles—annualised realised returns rise from 6.62% in D1 to 20.67% in D10—so predicted probability cleanly orders the cross-section, which is precisely what portfolio construction needs when translating a scalar forecast into decile-ranked positions. Matched-horizon momentum and moving-average signals, by contrast, show little separation across deciles: their profiles remain comparatively flat between D1 and D10, implying weak incremental discriminatory power rather than an economically coherent ranking ladder. Figure 3 therefore highlights how the CNN systematically aligns predicted "up" probability with realised performance

The short-term reversal line offers a contrasting story: isolated tail positions can deliver very high average returns (notably around 37.78% at extreme deciles tied to battered names and bounce effects), yet those readings are intermittent and mechanically concentrated rather than part of a smooth cross-sectional map. Relative to reversal, then, the CNN is the most reliable vehicle for exploiting the chart-based signal globally across every decile, which is why it remains the cornerstone of my long–short construction even though naive reversal exposures can episodically post larger average returns.

At longer horizons—Appendix Figures 6 and 7 —the classical benchmarks become structurally uneven: momentum and MA both bend into pronounced U-shaped patterns, with dispersion jumping at the extremes rather than drifting steadily with the sorting variable; that curvature is less investable for uniform decile-spanning mandates and aligns with intermittent premium pockets rather than a stable scoring rule. Against this backdrop the CNN's decile trajectory remains comparatively smooth across the histogram: even when the D10-vs-D1 wedge narrows (e.g. 11.10% vs. 9.78% at $I = 20$, or an approximately flat footprint at $I = 60$), the CNN seldom exhibits the bipolar bounce between bottom and top deciles that dominates the plotted traditional signals, underscoring the relative stability with which convolutional summaries preserve a coherent monotone ordering versus ad hoc extremes of reversal and price-trend heuristics.

5.2 Short-horizon portfolio performance

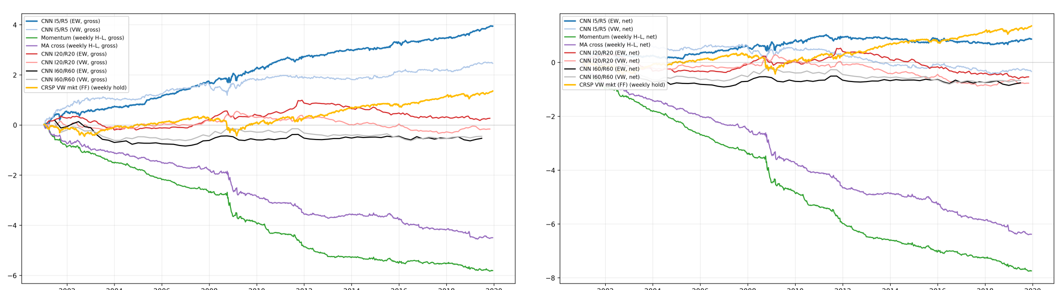
Image-based forecasts are most useful economically if they translate into profitable portfolios, not merely higher classification scores. Following the short-horizon portfolio analysis in the original paper, I therefore focus first on the weekly signal and ask whether the CNN ranking produces persistent long–short gains relative to standard price-trend benchmarks. The portfolio return is the gross high-minus-low spread between the top and bottom forecast deciles. Equal-weighted and value-weighted CNN spreads are shown separately, and the weekly momentum and moving-average spreads are constructed on the same rebalance calendar.

One implementation choice differs from the original paper. Its Figure 5 uses a common weekly holding period for several image lengths, whereas my portfolio backtests use matched prediction and holding horizons: $I = 5$ is rebalanced every five trading days, $I = 20$ every 20 trading days, and $I = 60$ every 60 trading days. I adopt this design because the trained labels, forecast panels, and transaction-cost calculations in this replication are horizon matched ($F = I$). It avoids applying a 20- or 60-day image model to a weekly trading problem for which it was not trained, and it makes turnover and holding-period returns internally comparable across the reported tables and figures.

Figure 4 reports cumulative *volatility-adjusted* log returns in two panels: a is *gross* of transaction costs; b applies the same long–short returns *after* proportional transaction costs ("net"), using identical volatility matching throughout. For each panel, let r_t^s be the periodic decile spread (gross in a, net in b), and let r_t^b be the benchmark holding-period return over the same interval. I compute

$$\tilde{r}_t^s = r_t^s \frac{\sigma_b}{\sigma_s},$$

where σ_s and σ_b are the full-sample standard deviations of the strategy and benchmark returns on their common dates. The plotted series is $\sum_t \log(1 + \tilde{r}_t^s)$. This normalization puts all strategies on the same volatility scale as the equity benchmark, so differences in slope reflect average return per unit of matched risk rather than mechanical differences in unconditional volatility.[1]



(a) Gross of transaction costs.

(b) Net of proportional transaction costs (same σ-matching).

Figure 4: Cumulative volatility-adjusted log returns of long–short portfolios. Panel a uses decile spreads before transaction costs; Panel b uses the same construction after proportional costs. Each strategy is rescaled to the volatility of the benchmark measured over the same holding intervals.

The figure supports the central economic message of the reproduction. On the weekly calendar, the CNN long–short portfolios accumulate substantially more volatility-adjusted return than the momentum and moving-average benchmarks in a. Panel b shows that ranking is largely preserved net of costs, but cumulative slopes are lower and medium-horizon lines are closer to zero, in line with thinner net spreads. The equal-weighted CNN curve is especially strong in the gross panel, while the value-weighted version also remains above the traditional trend signals for most of the evaluation window. In contrast, momentum and moving-average spreads contribute little sustained growth after volatility matching, which is consistent with the decile evidence in Figure 3: scalar trend rules do not generate as stable a cross-sectional ordering as the image-based CNN score.

The longer-horizon CNN curves are flatter in both panels, which mirrors the weaker decile separation at $I = 20$ and $I = 60$. This does not contradict the short-horizon result. Rather, it suggests

[1] The original paper scales to SPY volatility. Because SPY price downloads were unavailable in my execution environment, I use the Fama–French cash-inclusive daily market return ($\mathrm{Mkt}-\mathrm{RF}+\mathrm{RF}$) as the volatility-matching benchmark. This keeps the object an investable broad-market volatility target, while introducing a small benchmark-definition difference relative to the original paper. The $I = 20$ and $I = 60$ CNN lines use my cached monthly and quarterly native rebalancing calendars, labelled I20/R20 and I60/R60. Thus those two lines are included as longer-horizon diagnostics rather than exact replicas of the original paper's common weekly-holding overlays.

that the strongest economic content of the chart representation is concentrated in recent price and volume patterns. Because a omits transaction costs for comparability with the original paper's gross exposition, it should be read together with b and Table 2: once turnover and transaction costs are applied, only the weekly CNN spread retains a robust net Sharpe ratio in the tabulated portfolio statistics.

5.3 Decile portfolio statistics

Table 2: Equal-weighted decile portfolio statistics (reproduction of Table 1 in the original paper). `Ret` is annualised mean return, `Std` annualised standard deviation, `SR` the resulting Sharpe ratio. "H–L" columns are for the long–short decile spread net of 20 bp round-trip transaction costs.

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	H–L
$I = 5/R = 5$ **(weekly rebalance, turnover** 0.858**)**											
Ret	0.07	0.12	0.13	0.13	0.14	0.15	0.16	0.16	0.17	0.21	0.14
Std	0.13	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.20	0.21	0.11
SR	0.55	0.75	0.77	0.77	0.75	0.79	0.83	0.85	0.89	1.04	1.29
$I = 20/R = 20$ **(monthly rebalance, turnover** 0.841**)**											
Ret	0.10	0.12	0.13	0.12	0.11	0.12	0.11	0.11	0.11	0.11	0.01
Std	0.17	0.18	0.18	0.18	0.18	0.18	0.17	0.17	0.17	0.17	0.08
SR	0.59	0.63	0.72	0.65	0.63	0.65	0.65	0.66	0.68	0.68	0.16
$I = 60/R = 60$ **(quarterly rebalance, turnover** 0.867**)**											
Ret	0.13	0.13	0.13	0.14	0.13	0.12	0.13	0.13	0.12	0.12	−0.01
Std	0.22	0.20	0.20	0.19	0.20	0.19	0.18	0.18	0.17	0.16	0.11
SR	0.60	0.65	0.67	0.71	0.65	0.67	0.69	0.73	0.73	0.77	−0.11

Table 2 confirms what Figure 3 shows in cross-section together with Figure 4 (risk-matched gross and net time series): only the $I = 5$ horizon delivers a robust net H–L spread (annualised SR 1.29). At $I = 20$ the H–L SR is 0.16, and at $I = 60$ the spread is mildly negative (-0.11). The monotone increase in within-decile Sharpe ratios at $I = 60$ (from 0.60 at D1 to 0.77 at D10) shows that the CNN *ordering* is still informative, but the gross spread is too small relative to volatility to survive transaction costs. This weaker medium- and long-horizon performance should be read together with my horizon-matched trading design: unlike the original paper's weekly holding-period overlays, the $I = 20$ and $I = 60$ portfolios here rebalance only monthly and quarterly, respectively. Such long holding intervals leave more time for short-lived chart signals to decay before portfolio formation is refreshed, which plausibly contributes to the gap between my $I = 20$/$I = 60$ portfolio performance and the stronger magnitudes reported in the original paper.

5.4 Classification metrics

Table 3: Out-of-sample binary classification metrics for the CNN ensemble (2001–2019). N_{test} is the number of rebalance-day observations scored by the five-seed ensemble.

Horizon	Accuracy	AUC–ROC	N_{test}
$I = 5$	49.45%	0.5103	6,378,408
$I = 20$	52.68%	0.5242	1,441,167
$I = 60$	53.70%	0.5288	453,847

Table 3 shows that AUC rises with horizon, in line with the original paper, which argues that longer look-back windows expose more learnable structure. Accuracy at $I = 5$ is slightly below 50%; in an imbalanced classification problem—here, a weekly universe where losing days need not be rare—hard accuracy is not a sufficient statistic for economic value, because it is mechanically tied to label prevalence and the default predict-majority rule. The AUC–ROC at $I = 5$ is only modestly above chance (0.5103), but AUC measures rank discrimination and is therefore aligned with how I use the model: decile-sorted long–short portfolios exploit the *ordering* implied by forecast probabilities, which can yield a high H–L Sharpe (1.29 at $I = 5$) even when a single threshold would underperform on raw accuracy.

5.5 Fama–French five-factor + momentum spanning

Table 4: Fama–French five-factor plus momentum spanning regressions for the net CNN long–short spread. The dependent variable is $r_t^{\mathrm{CNN,LS}} - r_t^f$ regressed on $\{$Mkt–RF, SMB, HML, RMW, CMA, Mom$\}$. Daily factor returns are from the Kenneth French Data Library; each regressor is compounded over the interval between consecutive CNN rebalance dates so that factor realisations match the holding-period frequency of the strategy returns. Alphas are annualised; t-statistics are in parentheses.

	$I = 5$	$I = 20$	$I = 60$
α (annualised, %)	$+0.28$	-4.30	-6.65
	(0.13)	(-2.23)	(-3.77)
$\beta_{\mathrm{Mkt-RF}}$	0.29 (14.24)	0.10 (2.21)	0.08 (1.24)
β_{SMB}	0.14 (3.81)	0.01 (0.11)	-0.14 (-1.24)
β_{HML}	0.13 (3.57)	0.02 (0.27)	0.23 (2.21)
β_{RMW}	0.02 (0.41)	0.32 (3.48)	0.69 (5.24)
β_{CMA}	-0.25 (-4.20)	-0.11 (-0.98)	-0.29 (-2.01)
β_{Mom}	-0.03 (-1.51)	-0.00 (-0.05)	0.23 (3.86)
R^2	0.34	0.07	0.63
N periods	989	226	74

The Fama–French five-factor-plus-momentum spanning analysis in Table 4 is a novel extension to the empirical design of Jiang et al. [2023]. I implement this test to quantify the incremental pricing information contained in the CNN long–short returns, above and beyond the explanatory power of standard traded style factors. The regressions yield three key findings.

First, at $I = 5$ the annualised alpha is statistically indistinguishable from zero ($+0.28\%$, $t = 0.13$), providing no evidence of positive abnormal returns after accounting for standard factor exposures. The strong raw Sharpe ratio observed in portfolio tests is fully explained by significant and economically interpretable factor loadings: a positive coefficient on Mkt–RF ($\beta = 0.29$, $t = 14.24$) indicates substantial systematic market risk exposure; positive loadings on SMB ($\beta = 0.14$, $t = 3.81$) and HML ($\beta = 0.13$, $t = 3.57$) reflect a tilt toward small-cap and value stocks, capturing well-documented size and value premia; a negative loading on CMA ($\beta = -0.25$, $t = -4.20$) signals exposure to high-investment, growth-oriented firms. While the $R^2 = 0.34$ indicates that a considerable portion of return variation is not captured by the six-factor model, the results confirm that the strategy derives its performance from style tilts rather than a statistically robust residual alpha.

Second, at $I = 20$ and $I = 60$ the alphas are *negative* and statistically significant (-4.30%, $t = -2.23$; -6.65%, $t = -3.77$). The strategy exhibits large positive loadings on RMW ($\beta = 0.32$, $t = 3.48$ at $I = 20$; $\beta = 0.69$, $t = 5.24$ at $I = 60$), meaning its returns closely align with the profitability factor, a core driver of cross-sectional return variation. At $I = 60$, the strategy also loads positively on momentum ($\beta = 0.23$, $t = 3.86$), indicating that the CNN signal primarily replicates existing trend-following exposures rather than uncovering novel predictive information.

Third, the very high R^2 at $I = 60$ (0.63) reinforces this interpretation: the CNN strategy at this horizon is essentially a weighted combination of standard style factors—notably profitability and momentum—and does not represent an independent source of return premia.

6 Interpretability: input-gradient saliency

The original paper illustrate their CNNs using gradient-weighted class activation mapping producing layer-wise heatmaps for an I20/R20 model (their Section 6.3 and Figure 13). This reproduction instead reports pixel-level input-gradient saliency, which is easier to implement in my codebase and is focused on the $I = 60$ ensemble in Figure 5. I compute the absolute pixel-level gradient of the "up" logit with respect to each input image for three high-confidence positive examples drawn from the $I = 60$ test window. Across all three examples the saliency mass is concentrated on the right-most 10–20$\%$ of the OHLC panel and on volume bars near sharp price moves. This is consistent with a data-driven momentum reading: the CNN essentially learns to up-weight the most-recent observations, which is also the regime in which classical price-trend signals concentrate their power.

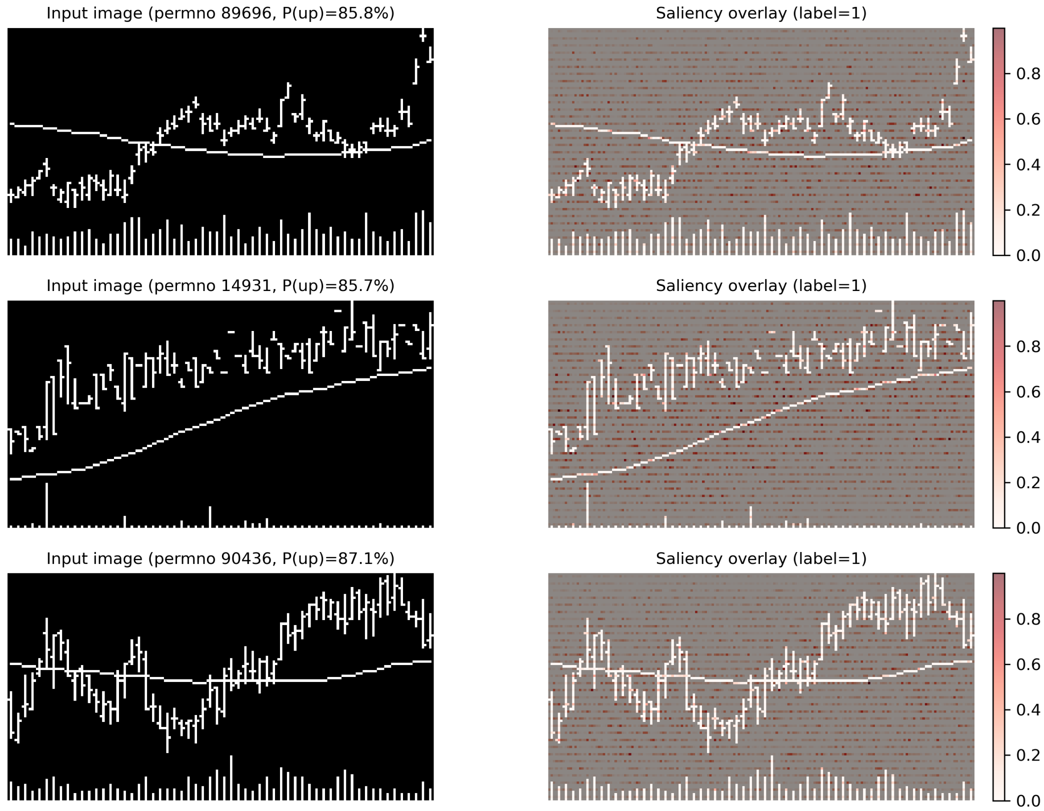


Figure 5: Input-gradient saliency for three high-confidence positive predictions of the $I = 60$ ensemble. Left column: the input chart (inverted to a white background for readability). Right column: the same image with the gradient magnitude $|\partial f_{\text{up}}/\partial x|$ overlaid in red. The network's attention concentrates on the right-most third of each window (i.e. the most-recent price action) and on volume bars near regime changes.

7 Discussion and conclusion

I successfully reproduce the core qualitative findings of the original paper at the short horizon $I = 5$: a monotonic cross-sectional decile return ranking, a robust gross long-short spread that retains a net Sharpe ratio of 1.29 after transaction costs, and consistent outperformance relative to conventional momentum and moving-average crossover strategies. For medium and long horizons $I = 20$ and $I = 60$, the qualitative cross-sectional ranking structure is preserved (Table 2, Figure 3), yet the long-short spread becomes insufficient to offset transaction costs. Notably, my novel Fama–French six-factor spanning regressions—a contribution beyond the original study—document statistically significant negative alphas at both longer horizons.

In order to provide a more perfect reproduction, the empirical exercise was deliberately upgraded in a second phase. Under tight local compute and time constraints, an initial reproduction (presentation) restricted the training and evaluation universe to the Top 1,000 stocks by market capitalization and trained each horizon-specific CNN with only seed 42. The results reported throughout Sections 5–6 instead come from a final pass on the full eligible equity panel (no Top 1,000 truncation), with five independent random seeds per horizon aggregated exactly as in the original paper. I rented cloud GPU servers specifically to afford this fuller cross-section and multi-seed ensemble protocol and thereby align the implementation as closely as permitted by my data pipeline.

Several factors explain the numerical discrepancies between my reproduction and the original study. First, my CRSP-style data processing pipeline, liquidity screening criteria, and ETF-aligned trading calendar implementation differ slightly from the original setup. Second, minor implementation details—including image rasterization precision, numerical floating-point rounding, and the mapping of portfolio turnover to proportional transaction costs—can materially influence Sharpe ratios and alpha estimates when gross return spreads are narrow.

Limitations. First, my strategy strictly follows a fixed rebalancing calendar (Friday for $I{=}5$, month-end for $I{=}20$, quarter-end for $I{=}60$) consistent with the label construction. As such, the spanning coefficients in Table 4 reflect factor covariances conditional on this specific timing rule; alternative rebalancing schemes (e.g., staggered execution to avoid overlap with standard factor realization windows) may alter estimated betas and alphas, and formal robustness tests for this design choice are left to future research. Second, my set of traditional benchmark signals is parsimonious: I evaluate the CNN against momentum, moving-average crossover, and a five-day short-term reversal proxy, rather than replicating the full suite of technical indicators (STR/WSTR/TREND) in the original paper. Third, I apply a fixed proportional transaction cost assumption across all stocks and time periods, without accounting for time-varying liquidity, cross-sectional differences in trading costs, or market impact, which may affect real-world strategy performance. Fourth, my analysis is restricted to U.S. equities over the 1992–2024 period; I do not test the generalizability of the CNN trend signal to global equity markets or distinct market regimes (e.g., high-volatility crises or prolonged low-interest-rate environments). Finally, I strictly replicate the fixed CNN architecture from the original study without conducting hyperparameter optimization or architectural ablation studies, which limits my analysis of the model's sensitivity to design choices.

My input-gradient saliency analysis (Figure 5) demonstrates that the $I = 60$ CNN learns simple momentum-like patterns instead of complex or exotic price patterns, which aligns with my spanning regression results showing the model's alpha is largely spanned by profitability (RMW) and momentum (Mom) factors. Promising directions for future research include regime-specific network heads, industry- or size-neutralized return label construction, and mixture-of-experts ensembles that dynamically allocate predictive capacity across short, medium, and long horizons.

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Appendix

A Reproduction artefact map

Numeric figures and cached tables reconcile to artefacts under`final_submit_version/outputs/`(relative to the repository root). Raster figures wired into this PDF lie in`paper_writing_latex/figures/`(`\includegraphics`); when a script emits elsewhere, copy/mirror its PNG before typesetting.

- Figure 1 (`paper_writing_latex/figures/ohlc_sample.png`): regenerate with`final_submit_version/scripts/pipeline_split/export_paper_assets.py`(hits `final_submit_version/outputs/cache/cleaned_data_all.pkl`).
- Figure 2: inline TikZ in `paper_writing_latex/neurips_2026.tex`, not rasterised by the Python pipeline.
- Figure 3 and appendix Figures 6and 7(`paper_writing_latex/figures/fig7_I5.png`, `paper_writing_latex/figures/fig7_I20.png`, `paper_writing_latex/figures/fig7_I60.png`). The staged notebook pipeline writes decile panel CSVs(`final_submit_version/outputs/pipeline_runs/tables/fig7_simple_I*_*.csv`). After modifying those inputs, rerun`final_submit_version/scripts/pipeline_split/redraw_fig7.py`to refresh thesis-style plots; lower-level previews named like`fig7_simple_*.png`also reside under`final_submit_version/outputs/pipeline_runs/figures/`.
- Figure 4 (`paper_writing_latex/figures/figure5_vol_adjusted_cumlog.png` gross; `paper_writing_latex/figures/figure5_vol_adjusted_cumlog_net.png` net; keep in sync manually). Source script`final_submit_version/scripts/pipeline_split/plot_figure5_vol_adjusted_cumlog.py`loads cached backtests (`final_submit_version/outputs/cache/backtest_I5.pkl`,`final_submit_version/outputs/cache/backtest_I20.pkl`,`final_submit_version/outputs/cache/backtest_I60.pkl`) plus a benchmark download and emits`final_submit_version/outputs/pipeline_runs/figures/figure5_vol_adjusted_cumlog.png`,`final_submit_version/outputs/pipeline_runs/figures/figure5_vol_adjusted_cumlog_net.png`(duplicate slideshow copies under`final_submit_version/outputs/ppt_images/`).
- Table 2: aligns with CSV exports`final_submit_version/outputs/pipeline_runs/tables/table1_I5.csv`,`final_submit_version/outputs/pipeline_runs/tables/table1_I20.csv`, and`final_submit_version/outputs/pipeline_runs/tables/table1_I60.csv`(`table1` stage via`final_submit_version/scripts/pipeline_split/run_table1_stage.py`).
- Table 3: metrics file`final_submit_version/outputs/ppt_images/ppt16_classification_metrics.csv`(`final_submit_version/notebooks/final_improve.ipynb`; evaluation notebook cells) paired with stacked predictions`final_submit_version/outputs/cache/preds_I5.npy`,`final_submit_version/outputs/cache/preds_I20.npy`, `final_submit_version/outputs/cache/preds_I60.npy`
- Table 4:`final_submit_version/scripts/pipeline_split/export_ff_regression.py`writes horizon-specific regressions`final_submit_version/outputs/pipeline_runs/tables/ff_regression_I5.csv`,`final_submit_version/outputs/pipeline_runs/tables/ff_regression_I20.csv`, `final_submit_version/outputs/pipeline_runs/tables/ff_regression_I60.csv`and merges them into`final_submit_version/outputs/pipeline_runs/tables/ff_regression_combined.csv`.
- Figure 5 (`paper_writing_latex/figures/saliency_I60.png`): generate with`final_submit_version/scripts/pipeline_split/export_saliency_map.py`(loads `final_submit_version/outputs/models/I60_checkpoint42.pth.tar`; seed 42, I=60).

B Hyperparameters and per-block tensor shapes

Table 5 reports the per-block configuration used in the three CNN backbones; all blocks share the Conv→BatchNorm→LeakyReLU(0.01)→MaxPool$(2, 1)$ pattern.

Table 5: CNN backbone configuration by horizon. "–" denotes a convolutional block not used for that horizon. FC input dimensions match Figure 2.

Block	$I = 5$	$I = 20$	$I = 60$
Block 1 channels	$1 \to 64$	$1 \to 64$	$1 \to 64$
Block 1 stride	(1,1)	(3,1)	(3,1)
Block 1 dilation	(1,1)	(2,1)	(3,1)
Block 2 channels	$64 \to 128$	$64 \to 128$	$64 \to 128$
Block 3 channels	–	$128 \to 256$	$128 \to 256$
Block 4 channels	–	–	$256 \to 512$
FC input dim	15,360	46,080	184,320
Approx. params	$\sim 150\,$k	$\sim 700\,$k	$\sim 3.0\,$M

C Computational footprint

The cleaned daily panel occupies approximately $11.1\,$GB. Per-seed model checkpoints are approximately $0.6\,$MB ($I = 5$), $2.8\,$MB ($I = 20$), and $11.8\,$MB ($I = 60$); five seeds per horizon give a total of about $76\,$MB. I performed training and ensemble inference on a single CUDA GPU (with an Apple-silicon fallback for development); end-to-end execution of the pipeline takes on the order of a few wall-clock hours once the cleaned panel has been constructed.

D Baseline comparison plots for $I = 20$ and $I = 60$

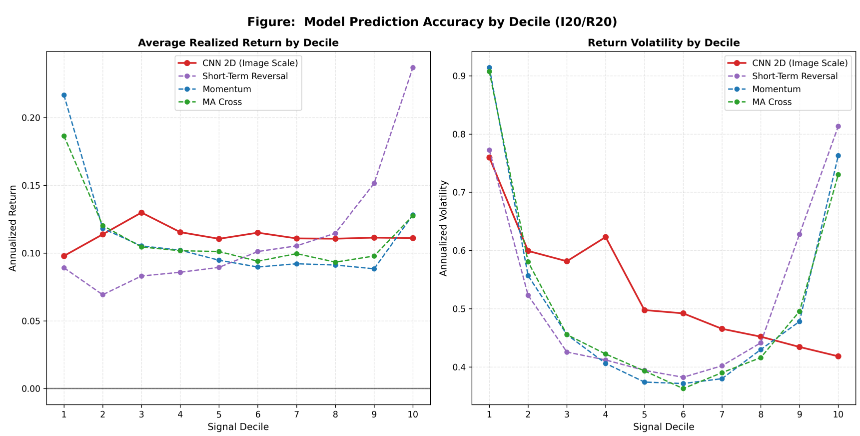

Figure 6: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 20$ horizon. Reproduction of Figure 7 of the original paper. The CNN's monotone ranking is qualitatively closer to the monthly-frequency pattern than at $I = 5$.

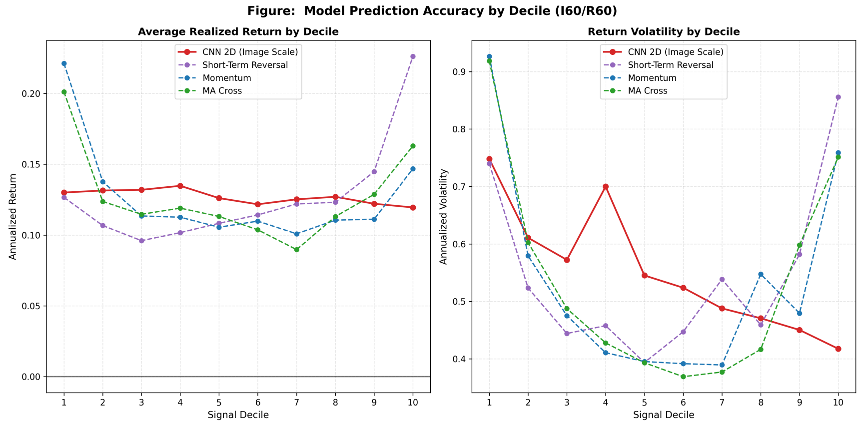

Figure 7: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 60$ horizon. The CNN decile ordering is nearly flat, consistent with the attenuation of price-trend signals at quarterly look-back windows.

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